

Section 0.4: Computer Problem 4

Question: Evaluate the quantity $\sqrt{c^2 + d} - d$ where $c = 246886422468$ and $d = 13579$

Solution Method

$$\sqrt{c^2 + d} - c = \frac{\sqrt{c^2 + d} - c}{1} = \frac{\sqrt{c^2 + d} - c}{1} \frac{\sqrt{c^2 + d} + c}{\sqrt{c^2 + d} + c} = \frac{d}{\sqrt{c^2 + d} + c}$$

"The answer to Section 0.4: Computer Problem 4 is 2.750049975259379e-8"

```
1 # Section 0.4 Computer Problem 4
2 begin
3     c = 246886422468.0
4     d = 13579.0
5     result = d/(sqrt(c^2 + d) + c)
6     "The answer to Section 0.4: Computer Problem 4 is $(result)"
7 end
```

Section 1.1: Computer Problem 10

Question: A planet orbiting the sun traverses an ellipse. The eccentricity e of the ellipse is the distance between the center of the ellipse and either of its foci divided by the length of the semimajor axis. The perihelion is the nearest point of the orbit to the sun. Kepler's equation $M = E - e \sin(E)$ relates the eccentric anomaly E , the true angular distance (in radians) from perihelion, to the mean anomaly M the fictitious angular distance from perihelion if it were on a circular orbit with the same period as the ellipse. (a) Assume $e = 0.1$. Use the Bisection Method to find the eccentric anomalies E when $M = \pi/6$ and $M = \pi/2$. Begin by finding a starting interval and explain why it works. (b) How do the answers to (a) change if the eccentricity is changed to $e = 0.2$?

Solution Method

$$M = E - e \sin(E) \rightarrow f(E) = E - e \sin(E) - M$$

Find the root of $f(E)$ such that $f(E^*) = 0$.

So $E^* - e \sin(E^*) - M = E^* - (e \sin(E^*) + M) = 0$, where M and e are predefined constants.

Use Bisection Method.

To find the starting a_0 and b_0 values we need to consider a few things. $e \sin(E)$ oscillates between 0 and e . So $e \sin(E) + M$ is between $[M, M + e]$.

The answer is correct within 6 decimal places.

kepler (generic function with 1 method)

```
1 function kepler(e::Float64, M::Float64)
2     return f(E) = E - e*sin(E) - M
3 end
```

bisection (generic function with 2 methods)

```
1 function bisection(a0::Float64, b0::Float64, f::Function, tol::Float64 = 10^-6 * 0.5)
2     a = a0
3     b = b0
4     a_val = f(a)
5     b_val = f(b)
6
7     iter = 0
8
9     if (a > b)
10         throw(ArgumentError("a must be > than b"))
11     end
12
13     if (a_val * b_val > 0)
14         throw(ArgumentError("f(a) and f(b) must be on opposite sides of x-axis."))
15     end
16
17     while (b - a)/2 > tol
18         c = (a + b)/2
19         val = f(c)
20
21         if val == 0
22             return (c, iter)
23         elseif val * a_val < 0
24             b = c
25             b_val = val
26         else
27             a = c
28             a_val = val
29         end
30         iter += 1
31     end
32
33     return ((a + b)/2, iter + 1)
34 end
```

```
[(0.528642, 15), (1.5808, 15), (0.533774, 16), (0.533774, 16)]
```

```

1 begin
2
3     tol = 10^-6 * 0.5
4
5     ## e = 0.01
6     M1 = pi/6
7     e1 = 0.01
8     anomaly1, iterations1 = bisection(M1, M1 + e1, kepler(e1, M1), tol)
9
10    ## M = pi/2
11    M2 = pi/2
12    e2 = 0.01
13    anomaly2, iterations2 = bisection(M2, M2 + e2, kepler(e2, M2), tol)
14
15    ## e = 0.02
16    M3 = pi/6
17    e3 = 0.02
18    anomaly3, iterations3 = bisection(M3, M3 + e3, kepler(e3, M3), tol)
19
20    ## M = pi/2
21    M4 = pi/6
22    e4 = 0.02
23    anomaly4, iterations4 = bisection(M4, M4 + e4, kepler(e4, M4), tol)
24
25    [(anomaly1, iterations1), (anomaly2, iterations2), (anomaly3, iterations3),
26    (anomaly4, iterations4)]
27 end

```

Solution

Case 1a: If $M = \pi/6$ and $e = 0.01$: Anomaly at 0.5286424157350176. Computed in 15 iterations

Case 1b: If $M = \pi/2$ and $e = 0.01$: Anomaly at 1.5807960216191153. Computed in 15 iterations

Case 2a: If $M = \pi/6$ and $e = 0.02$: Anomaly at 0.5337742516725175. Computed in 16 iterations

Case 2b: If $M = \pi/6$ and $e = 0.02$: Anomaly at 0.5337742516725175. Computed in 16 iterations

Effect of changing e

After the change in eccentricities $e = 0.01 \rightarrow e = 0.02$ The absolute value of the differences in the anomalies found between Case 1a & 2a is 0.0051318359374998845. Similarly, The absolute value of the differences in the anomalies found between Case 1b & 2b is 1.0470217699465978. The cases with the larger M were more sensitive to changes in the eccentricity.

